



HVAC Components

Compressors · Condensers · Expansion · Evaporators · AHUs

• HVAC ENGINEERING • LESSON 03

The Four Core Refrigeration Components

Every vapour-compression HVAC system — from a window AC to a large chiller — is built around the same four components: **compressor, condenser, expansion device, and evaporator**. These four create the refrigeration cycle you studied in HVAC-002. This lesson breaks down each component in depth, then covers the air-side equipment (AHUs, fans, filters) that delivers conditioned air to occupied spaces.

The Compressor — Heart of the System

The compressor is the only powered component in the refrigeration cycle. It draws in low-pressure refrigerant vapour from the evaporator and compresses it to high pressure and temperature, ready to reject heat at the condenser. Compressor type dictates system size, efficiency, noise, and serviceability.



Figure 1 — Compressor Types and Applications

Type	Capacity Range	COP Range	Typical Use
Scroll	1–30 kW	3.0–4.5	Residential split systems, small VRF units
Reciprocating	0.5–40 kW	2.5–3.5	Small commercial, refrigeration, packaged units
Rotary	0.5–5 kW	2.5–3.2	Window AC, portable AC, small inverter splits
Twin-screw	50–1 500 kW	4.5–6.5	Industrial chillers, large commercial buildings
Centrifugal	500–10 000 kW	5.0–7.0	District cooling, large hospitals, data centres

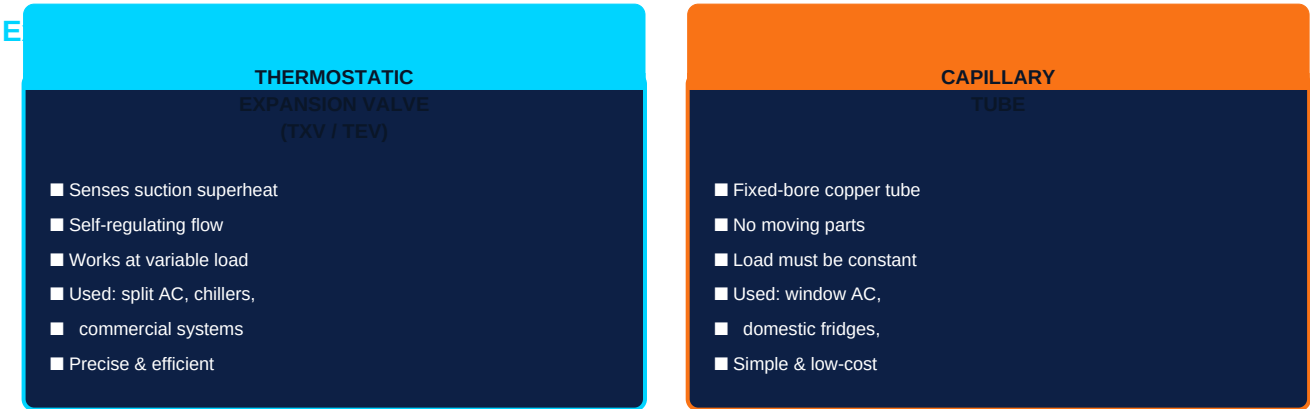
Condenser — Heat Rejection to Environment

Type	Heat Rejection Method	Efficiency	Typical Location
Air-Cooled Condenser	Ambient air blown over refrigerant coil by a fan	Lower — limited by outdoor dry-bulb temperature	Rooftop, side wall, outdoor unit
Water-Cooled Condenser	Water circulates through shell-and-tube or plate HX	Higher — limited by wet-bulb temperature (cooling tower)	Plant room; paired with cooling tower or water loop
Evaporative Condenser	Water sprayed over coil; evaporation removes heat	Between air- and water-cooled	Industrial plants, supermarket refrigeration

Rule of thumb: Water-cooled condensers are 15–25% more efficient than air-cooled equivalents but require a cooling tower, water treatment, and a dedicated plant room. Air-cooled is standard for buildings under ~500 kW.

Expansion Devices and Evaporators

After the condenser, high-pressure liquid refrigerant passes through an expansion device that drops its pressure and temperature sharply — creating the cold, low-pressure mix that absorbs heat in the evaporator.



Air-Side Equipment: AHU, Fans and Filters

The refrigeration components handle the *thermal* side of an HVAC system. The air-side equipment handles *distribution* — moving conditioned air from the plant to every room in the building. The Air Handling Unit (AHU) is the central hub of the air-side system.

The Air Handling Unit (AHU)

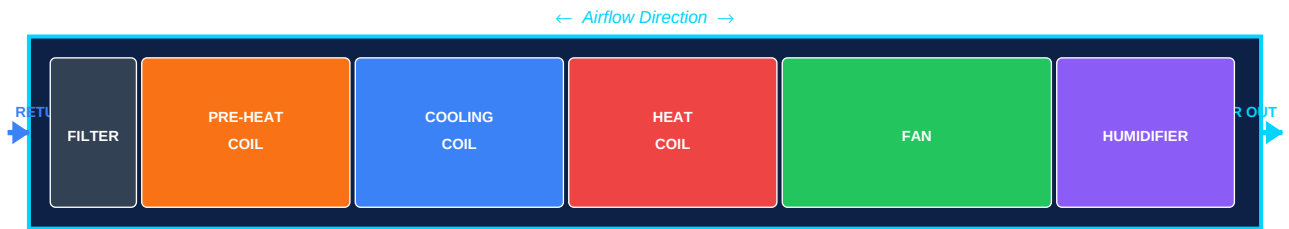


Figure 2 — Air Handling Unit (AHU) Component Layout

AHU Section	Function	Key Specification
Pre-filter (G4/M5)	Removes coarse dust, fibres, insects from return or fresh air	Efficiency class: ISO Coarse/Medium; replace at ΔP limit
Heating Coil (Pre-heat)	Heats incoming fresh air in winter before mixing with return air	LTHW (70/50°C) or electric; sized for minimum outdoor temp
Cooling Coil (DX or CHW)	Cools and dehumidifies air to supply set-point (typically 12–14°C)	CHW (6/12°C) or DX; approach temp 2–3 K; drain pan required
Re-heat Coil	Raises supply air temp after cooling to avoid overcooling rooms	LTHW, electric, or heat recovery; active in swing seasons
Supply Fan	Draws air through AHU and delivers it to ductwork at design pressure	Centrifugal forward-curved or backward-curved; VSD for efficiency
Humidifier	Adds moisture in winter when outdoor air is too dry (RH < 30%)	Steam lance, evaporative pad, or ultrasonic; controlled by RH sensor
HEPA / F9 Filter	High-efficiency final filter for hospitals, labs, clean rooms	ISO ePM1 ≥ 85%; high pressure drop — requires larger fan

Fan Types in HVAC

Fan Type	Flow Pattern	Pressure	Best For
Centrifugal — Forward Curved	Radial outward	Medium	Low-resistance AHU duty; quiet at low speed
Centrifugal — Backward Curved	Radial outward	Medium–High	High-efficiency AHU; stable characteristic curve
Plug / EC Fan	Radial	Medium–High	Plug-fan AHUs; EC motor gives 20–35% energy saving
Axial / Propeller	Axial	Low	Condenser fans, cooling towers, wall extract fans
Mixed Flow	Diagonal	Medium	Inline duct fans where low noise and good pressure needed

What comes next — HVAC-004: Air Conditioning Basics — how split systems, VRF/VRV multi-splits, and packaged rooftop units combine these components into complete, installed systems.